Jon Glenn Gjevestad: Multiple T-test for quality control

Except for part 7, this is the “Multippel T-test” paper from Jon Glenn Gjevestad translated by Trond Arve Haakonsen. See example exercise no. 9 (2018) in course TBA4245 Geodesy for a numerical example of the multiple T-test.

Summary

In the theory of *least squares method (LSM)* it is common to assume that the observation errors are uncorrelated identically distributed. If this is the case, the LSM estimator gives an optimal solution based on minimum *Sum of Squared Errors (SSE).*

But if one blunder/outlier/gross error is hidden among the observations, the use of LSM will cause a propagation from the blunder to all residuals. This makes the blunder very difficult to detect, and the assumption of normally distributed observational errors are no longer valid.

Therefore it is developed methods for detection and identification of blunders among observations before final adjustment is made. Examples are:

1 Datasnooping (Baarda)
2 Tau test
3 Multiple *t*-test

In this paper a multiple *t*-test is selected as the basis for quality control of observations. A commonly used three steps control procedure is:

1 Detection (Test of SSE)
2 Identification (Multiple *t*-test)
3 Adaption (Observation of largest *t*-value removed from the system)

1 Test of SSE

This test is one of more so-called global tests to handle the entire system. The test will give an indication of whether it is constraints in the system, and is often used for detection of blunders. The hypotheses for testing SSE are:

$$H_0: \quad \hat{\sigma}_0^2 = \sigma_0^2$$
$$H_1: \quad \hat{\sigma}_0^2 \neq \sigma_0^2$$

where σ_0 is a priori standard deviation of weight unit, and $\hat{\sigma}_0$ the a posteriori standard deviation at the weight unit. The test statistics in this case is:

$$z = \frac{\mathbf{v}^{\mathrm{T}}\mathbf{P}\mathbf{v}}{\sigma_0^2} = \frac{\hat{\sigma}_0^2}{\sigma_0^2}(n-e) \sim \chi^2_{(n-e)} \tag{1}$$

where **P** is the weight matrix and **v** is the residual vector (estimated corrections). *n* is the number of observations and *e* is the number of unknowns. If $\chi^2_{(n-e,\alpha/2)} < z < \chi^2_{(n-e,1-\alpha/2)}$ the zero hypothesis is accepted, and it cannot be claimed that $\hat{\sigma}_0^2$ is significant different from σ_0^2.

Sometimes it is necessary to compare SSE's from two adjustments (e.g. constraint- and minimum constraint, or to "set one known point free"). The hypotheses for testing differences or changes in SSE's are:

$$H_0: \quad \mathbf{v}_2^{\mathbf{T}}\mathbf{P}\mathbf{v}_2 = \mathbf{v}_1^{\mathbf{T}}\mathbf{P}\mathbf{v}_1$$
$$H_1: \quad \mathbf{v}_2^{\mathbf{T}}\mathbf{P}\mathbf{v}_2 > \mathbf{v}_1^{\mathbf{T}}\mathbf{P}\mathbf{v}_1$$

The test statistics in this case is:

$$f = \frac{(\mathbf{v}_2^{\mathbf{T}}\mathbf{P}\mathbf{v}_2 - \mathbf{v}_1^{\mathbf{T}}\mathbf{P}\mathbf{v}_1)/(\nu_2 - \nu_1)}{\mathbf{v}_1^{\mathbf{T}}\mathbf{P}\mathbf{v}_1/\nu_1} \sim F_{(\nu_2-\nu_1,\nu_1)} \qquad \begin{array}{l} \nu_2 = n_2 - e_2 \text{ and } \nu_1 = n_1 - e_1 \\ \text{are the degrees of freedom} \end{array} \tag{2}$$

If $f < F_{((\nu_2-\nu_1),\nu_1,\alpha)}$ the zero hypothesis cannot be rejected, so we cannot claim that $\mathbf{v}_2^{\mathbf{T}}\mathbf{P}\mathbf{v}_2$ is significant different from $\mathbf{v}_1^{\mathbf{T}}\mathbf{P}\mathbf{v}_1$.

(Note1: different degrees of freedom than Gjevestads Norw. ver. Note2: The global test can also be performed after the blunder estimation in ch. 2 (gross error test), to check the quality of the known coordinates and heights.)

2 Blunder estimation

Adding an extra column vector with zero elements, except for value one at element *i* to the design matrix, a blunder can be estimated simultaneously with unknown parameters.

$$\mathbf{A}_{\mathbf{i}} = \begin{bmatrix} \mathbf{A} & \mathbf{e}_{\mathbf{i}} \end{bmatrix} \tag{3}$$

Index *i* means that we are testing for a blunder in observation *i*. As an example, $\mathbf{e}_1^{\mathbf{T}} = \begin{bmatrix} 1 & 0 & 0 & . & 0 \end{bmatrix}$ is used to estimate a blunder in observation no. 1.

$$\mathbf{A}_1 = \begin{bmatrix} - & - & . & 1 \\ - & - & . & 0 \\ . & . & . & . \\ - & - & . & 0 \end{bmatrix} \tag{4}$$

The blunder ∇_i is estimated simultaneously with the unknown parameters as:

$$\hat{\mathbf{x}}_{\mathbf{i}} = (\mathbf{A}_{\mathbf{i}}^{\mathbf{T}}\mathbf{P}\mathbf{A}_{\mathbf{i}})^{-1}\mathbf{A}_{\mathbf{i}}^{\mathbf{T}}\mathbf{P}\mathbf{l} \qquad \text{where} \qquad \hat{\mathbf{x}}_{\mathbf{i}}^{\mathbf{T}} = \begin{bmatrix} - & - & . & \nabla_i \end{bmatrix} \tag{5}$$

The co-factor matrix of the unknown parameters: $\mathbf{Q}_{\mathbf{xx}} = (\mathbf{A}_{\mathbf{i}}^{\mathbf{T}}\mathbf{P}\mathbf{A}_{\mathbf{i}})^{-1} = \begin{bmatrix} - & - & . & - \\ - & - & . & - \\ . & . & . & . \\ - & - & . & q_{\nabla\nabla} \end{bmatrix}$ (6)

with the blunders co-factor as the last element on the diagonal.

A posteriori variance for the weight unit, without influence from the blunder ∇_i is estimated (for each i) as:

$$\hat{\sigma}_0^2 = \frac{\mathbf{v}_i^{\mathbf{T}} \mathbf{P} \mathbf{v}_i}{(n-e-1)} \tag{8}$$

Note the loss of one degree of freedom because one observation has been used to estimate the blunder. (Tronds comment: ensures "delete-one" estimates. A blunder in obs. i is adopted in the model sum of squares, and do not contribute to the SSE that is used to estimate $\hat{\sigma}_0^2$).
The standard deviation for the estimated gross error can now be calculated as:

$$\hat{\sigma}_{\nabla} = \hat{\sigma}_0 \sqrt{q_{\nabla\nabla}} \tag{9}$$

3 Search for blunders (Multiple T-test, minimum constraint)

The influence at the result from a blunder depends on the geometry of the system, and the weights. The hypotheses are:

$$H_0 : \quad \nabla_i = 0$$
$$H_1 : \quad \nabla_i \neq 0$$

where ∇_i represents a blunder in observation i. The test statistics for a multiple t-test is:

$$t = \frac{\hat{\nabla}}{\hat{\sigma}_{\nabla}} \sim T_{(n-e-1)} \tag{10}$$

If $t > T_{(n-e-1,1-\alpha/2)}$ the zero hypothesis will be rejected and one can claim: $\nabla_i \neq 0$. That means a probable blunder is identified in obs i.

The observation with the largest t-value is then removed from the system (adaptation) and the gross error search is repeated until all observations have passed the test.

4 Internal reliability (minimum constraint)

The redundancy number of an observation explains the proportion of a possible gross error that appears in the observation residual. The redundancy number for observation i is the diagonal element r_{ii} in the **R-**matrix given as:

$$\mathbf{R} = \mathbf{Q}_{\mathbf{vv}} \mathbf{P} \tag{11}$$

$$\mathbf{Q}_{\mathbf{vv}} = \mathbf{P}^{-1} - \mathbf{A} \mathbf{Q}_{\mathbf{xx}} \mathbf{A}^{\mathbf{T}} \tag{12}$$

For uncorrelated observations we have $0 < r_{ii} < 1$.

The Largest Remaining gross Error (LRE) is that particular end point of a confidence interval around the estimated blunder, which has maximum numerical value.

$$\nabla_{\max_i} = \max(|\hat{\nabla}_i \pm \hat{\sigma}_{\nabla} \cdot T_{(n-e-1,1-\alpha/2)}|) \tag{13}$$

Internal reliability explains how large the blunders can be, with the given probability for correct decision.

5 External reliability (constraint adjustment)

External reliability explains how large damage the LRE can make at the unknowns or a function of them, for example the coordinates.

$$\nabla_{\mathbf{x}} = (\mathbf{A}^{\mathrm{T}}\mathbf{P}\mathbf{A})^{-1}\mathbf{A}^{\mathrm{T}}\mathbf{P}\nabla_{max} \tag{14}$$

$\nabla_{\mathbf{x}}$ is the effect at estimated parameters from ∇_{max_i}, where ∇_{max} is a column vector of zeroes, except for the LRE in the i^{th} element: $\nabla^T_{max} = \begin{bmatrix} 0 & 0 & . & \nabla_{max_i} & . & 0 \end{bmatrix}$.

6 Bonferroni correction

For multiple comparisons in one data set, the significance level should be corrected to take into account the increased probability for the Type I error (i.e. the probability rejecting the null hypothesis when it is true). The probability of not making a Type I error making n independent multiple tests can be written as:

$$(1-\alpha)\cdot(1-\alpha)\ldots = (1-\alpha)^n \tag{15}$$

We distinguish between two meanings of significance level *α:*

- α_{pt} the probability of making a Type- I error per test.
- α_{pf} the probability of making a Type I error per family of tests.

Thus the probability of making at least one Type I error for multiple tests can be written as:

$$\alpha_{pf} = 1-(1-\alpha_{pt})^n \tag{16}$$

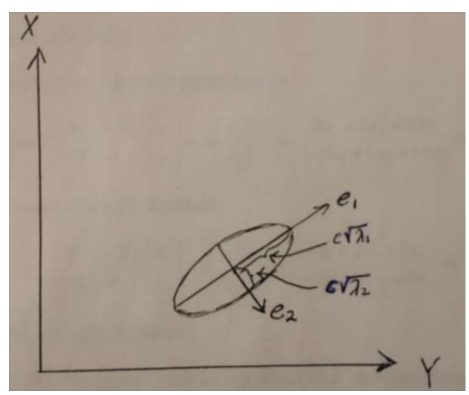
Rearranging (16) give:

$$\alpha_{pt} = 1-(1-\alpha_{pf})^{1/n} \tag{17}$$

For large n this term is approximated by the linear part of the Taylor series expansion of equation (17) only, and is then called the Bonferroni correction:

$$\alpha_{pt} \approx \alpha_{pf} / n \tag{18}$$

7 Confidence ellipses (Not translated)



An alternative to Gjevestads equations is to compute the eigenvalues and normalized eigenvectors of the cov matrix Σ_{xx}.

Matlab: `[ E, Lam ] = eig(Cxx)`

- The length of semi axes in a standard error ellipse is the square root of the diagonal elements in Lam (λ)
- The direction of the ellipse axes is identical to the eigen vectors directions.

c is a scale factor to compute semi axes lengths for uncertainties different from the standard *39.3%* ellipse ($c=1$). For a covariance matrix from estimation (unknown variance, see compendia) we find the 95 persent ellipse using $c = 2.45$.